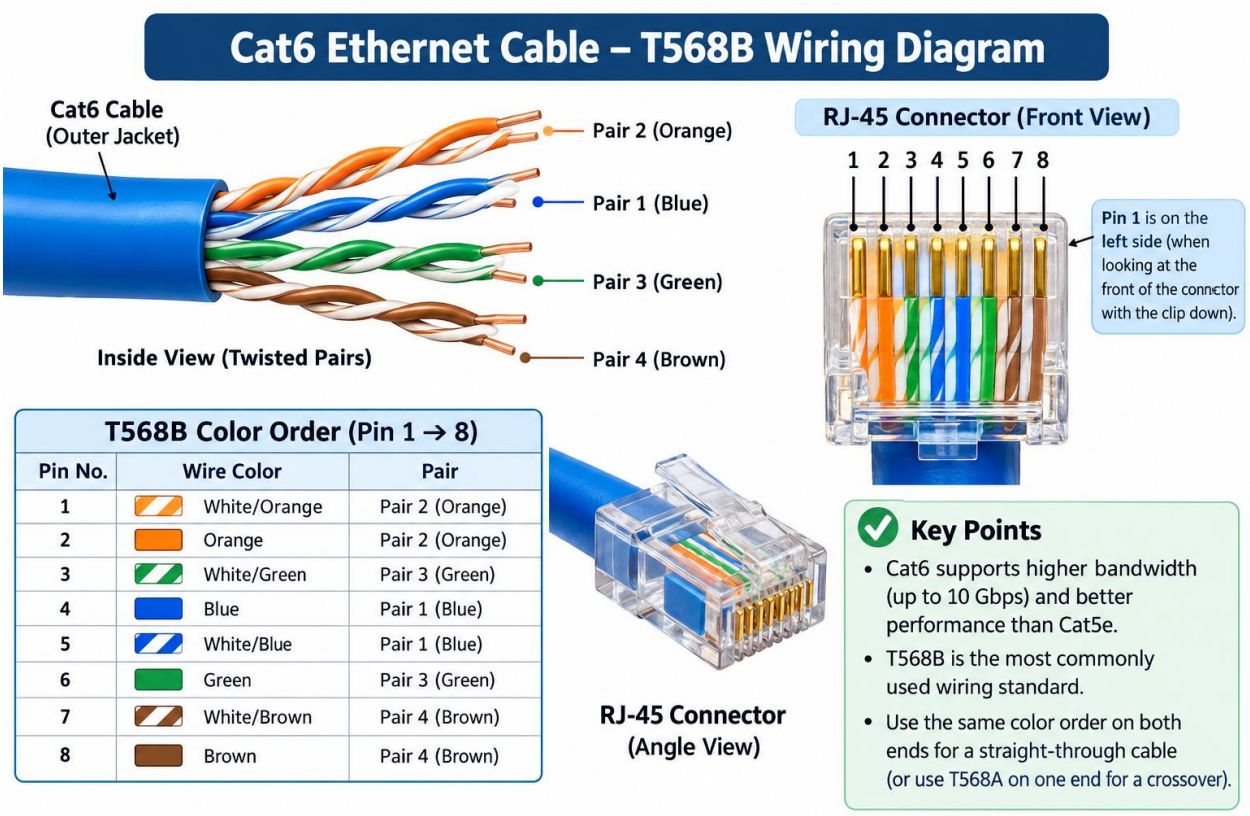
1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Ans:-



2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Ans:-

No.	Network Cable Type	Real-World Device/App	Why It Is Used
1	Cat5e	Wi-Fi Router	Connects a router to a computer or other network device at speeds up to 1 Gbps.
2	Cat6	Zomato POS Terminal	Provides a fast, reliable wired connection for processing restaurant orders and updating payments.
3	Fiber Optic	IPL Streaming Server	Supports high-speed data transfer over long distances for smooth live video streaming.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Ans:-

Tools required: Cat6 Ethernet cable, RJ-45 connector, crimping tool, and cable stripper.

1. Strip the cable: Remove about 2–3 cm of the outer jacket without damaging the inner wires.
2. Arrange the wires: Untwist and straighten the eight wires in the T568B color order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
3. Trim the wires: Cut the wires evenly so they fit inside the RJ-45 connector.
4. Insert the wires: Push all eight wires into the connector, ensuring each wire reaches the front and stays in the correct pin order.
5. Crimp the connector: Place the connector into the crimping tool and squeeze firmly to secure the wires.
6. Test the cable: Use a cable tester to check that all eight wires are connected correctly.

YouTube video: How to Crimp an RJ45 Connector – T568B Tutorial

Two common mistakes to avoid

1. Wrong wire order: Always follow the T568B color sequence correctly.
2. Loose connections: Ensure all eight wires reach the end of the connector and the cable jacket is secured before crimping.

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

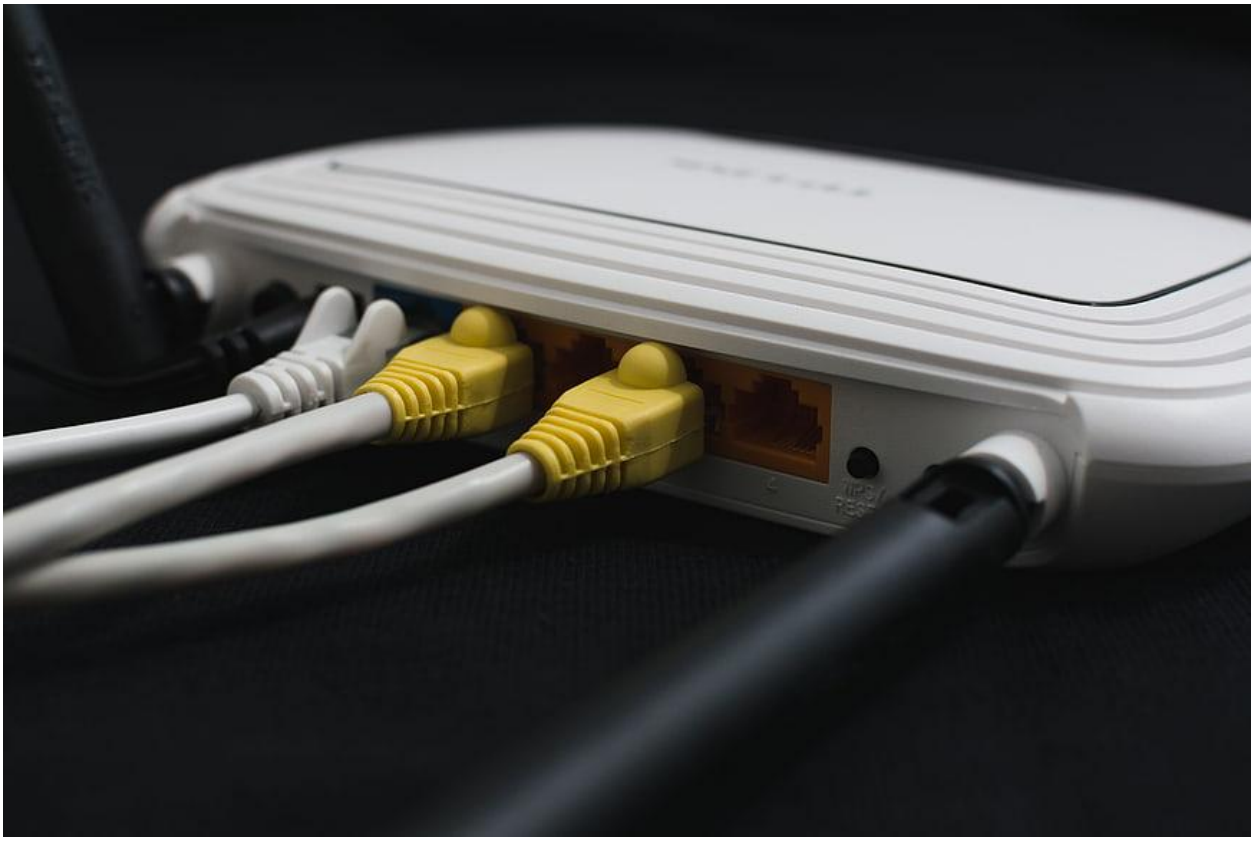
Ans:-

Connection	Cable	Connector	Why
Consoles / gaming PCs to the network switch	Cat6 (or Cat6a) Ethernet	RJ-45	Provides a wired gigabit connection, low ping, and fewer disconnections during live matches.
Main switch to a second switch / router (long distance or backbone)	Fiber-optic cable	LC connector (through SFP module)	Provides high speed over long distances and resists electromagnetic interference.
Console / media player to each screen	High-Speed HDMI (2.0 / 2.1)	HDMI	Carries high-quality video and audio together in one cable, supporting 4K when the devices and cable are compatible.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

Ans:-



Description:

1. Cable type: Ethernet LAN cable (Cat5e/Cat6).
2. Connector: RJ-45 connector, with a transparent plastic end.
3. Devices connected: Wi-Fi router and a laptop, desktop computer, or network switch.
4. Purpose: It transfers data between network devices and provides a wired network or internet connection.